Practical 3 — Association and group comparisons

Historical mutual-fund returns · 307312 Applied Statistics — Masters Level

- This practical accompanies measures of association and comparative boxplots in Part 2.
- Complete categorical data and numerical data first.
- Allow **60–75 minutes**, or two shorter sessions.

By the end, you should be able to:

- Compare groups using their distributions.
- Inspect a numerical relationship with a scatterplot.
- Calculate Pearson correlation.
- Distinguish an observed association from a causal explanation.

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1. The business questions

Return to the historical list of **259 mutual funds**:

> - How do Growth and Value funds compare on five-year annualized returns?
> - Is expense ratio associated with those returns in this list?

- An analyst needs distributions and numerical summaries, rather than a single ranking.
- The five-year return covers **1997–2001** and is expressed as an annualized percentage.
- For example, a recorded value of 10 means 10% per year over that period; it is not a total five-year return of 10%.

Variable	Meaning and unit
`objective`	Growth or Value
`expense_ratio_pct`	Expenses as a percentage of average net assets, as defined by the book
`return_5yr_pct`	Annualized return for 1997–2001, in percent
`return_2001_pct`	Twelve-month return during 2001, in percent
`risk`	Historical ordered risk classification

- A difference between two percentage returns is measured in **percentage points**.
- A single recorded return per fund does not give us annual observations for that fund or its year-to-year volatility.

2. Import and inspect

```
# This import cell works from the project root or the notebook folder.
data_dir <- if (dir.exists("data")) "data" else file.path("..", "data")
funds <- read.csv(file.path(data_dir, "practicals/funds2002.csv"),
                  stringsAsFactors = FALSE)
suppressPackageStartupMessages(library(dplyr))
source(file.path(data_dir, "..", "practicals", "_includes", "petra-plots.R"))
funds$objective <- factor(funds$objective, levels = c("Growth", "Value"))
funds$risk <- factor(funds$risk,
                     levels = c("very low", "low", "average", "high", "very
        high"),
                     ordered = TRUE)

# Check group sizes, missing values and numerical ranges before comparing results.
table(funds$objective)
```

```
Growth  Value
   158    101
```

```
colSums(is.na(funds[c("objective", "expense_ratio_pct", "return_5yr_pct")]))
```

```
       objective expense_ratio_pct    return_5yr_pct
               0                 0                 0
```

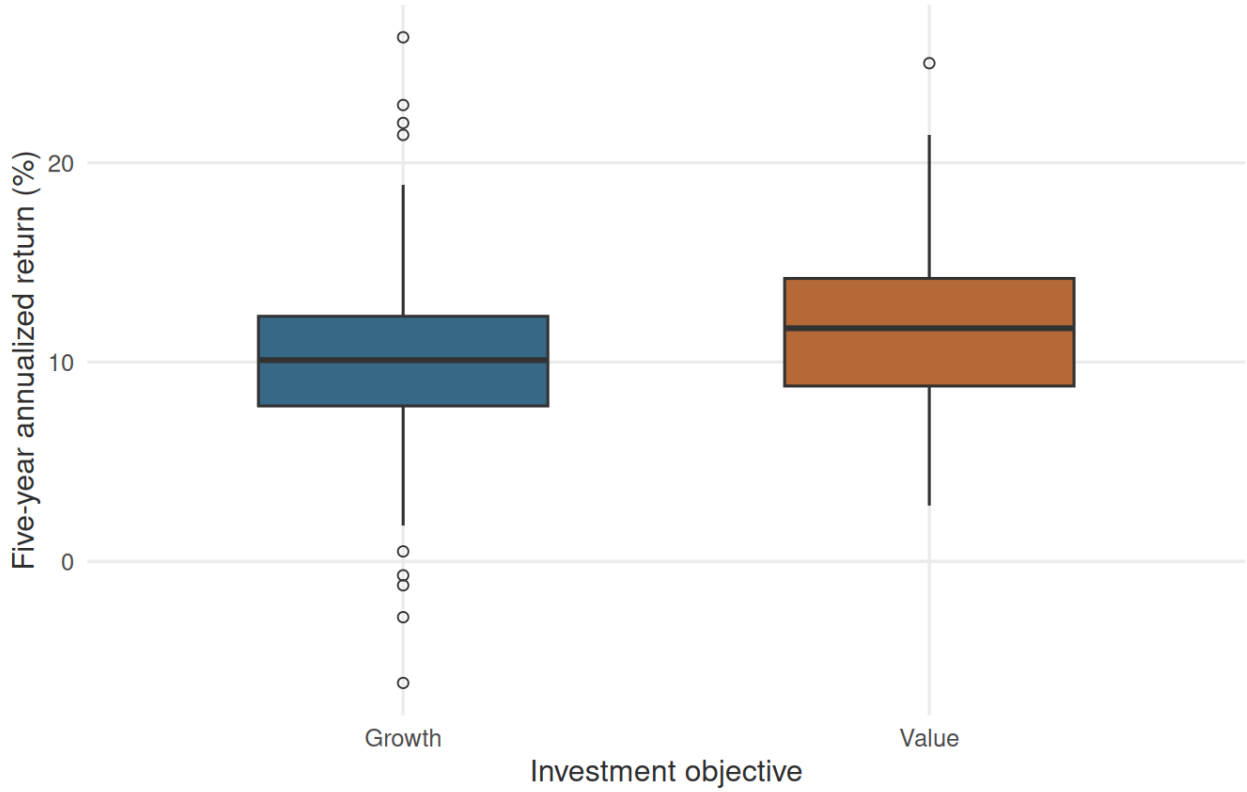
```
summary(funds[c("expense_ratio_pct", "return_5yr_pct")])
```

```
expense_ratio_pct return_5yr_pct
Min.   :0.110     Min.   :-6.10
1st Qu.:0.850     1st Qu.: 8.20
Median :0.980     Median :10.50
Mean   :1.042     Mean   :10.75
3rd Qu.:1.195     3rd Qu.:12.95
Max.   :2.130     Max.   :26.30
```

- `factor()` fixes a consistent display order.
- `table()` reports 158 Growth and 101 Value funds.
- `summary()` helps spot unexpected ranges; unusual values require investigation rather than automatic deletion.
- Download the CSV.

3. Compare distributions before calculating differences

```
# Boxplots show median, middle-half spread and observations beyond the whiskers.
ggplot(funds, aes(x = objective, y = return_5yr_pct, fill = objective)) +
  geom_boxplot(width = 0.55, outlier.shape = 1) +
  scale_fill_manual(values = petra_groups, guide = "none") +
  labs(x = "Investment objective", y = "Five-year annualized return (%)")
```



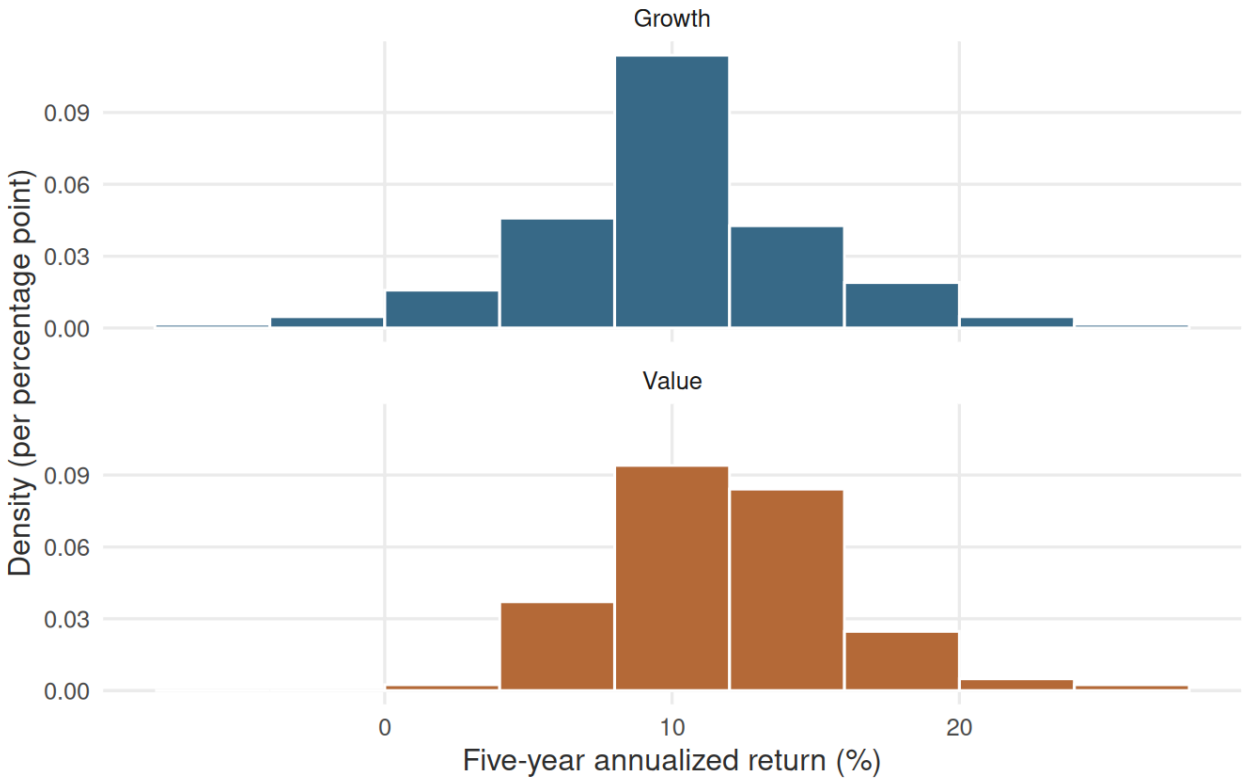
Five-year annualized returns by objective, 1997–2001.

- The line inside each box is the median.
- The box spans the middle half of observations using the plotting convention.
- Whiskers extend to observed values within 1.5 IQR of the hinges; points beyond them are flags for inspection, not necessarily errors.

Pause

- Which group appears to have the higher median?
- Which has more spread?
- Do the distributions overlap?
- Can a higher group median tell us that every fund in that group outperformed every fund in the other group?
- For a second view, compare histograms using a common bin width and horizontal scale.
- Each panel has its own group denominator.

```
# Density rescales unequal group sizes; each panel has total bar area 1.
ggplot(funds, aes(x = return_5yr_pct, y = after_stat(density), fill = objective))
        +
  geom_histogram(binwidth = 4, boundary = 0, colour = "white") +
  facet_wrap(~ objective, ncol = 1) +
  scale_fill_manual(values = petra_groups, guide = "none") +
  labs(x = "Five-year annualized return (%)", y = "Density (per percentage
        point)")
```



Relative distributions with a common 4-percentage-point bin width.

- Density height is not itself a percentage.
- Each bar's area is its group's relative frequency in that interval.
- Common bins make the displays comparable; the plot does not weight a fund by its assets.

4. Summarize the comparison

```
# group_by() separates objectives; summarise() computes the same statistics for
        each.
group_summary <- funds |>
  group_by(objective) |>
  summarise(n = n(), mean_pct = mean(return_5yr_pct),
            median_pct = median(return_5yr_pct), sd_pp = sd(return_5yr_pct),
            q1_pct = quantile(return_5yr_pct, 0.25),
            q3_pct = quantile(return_5yr_pct, 0.75), .groups = "drop")
knitr::kable(group_summary, digits = 2,
             col.names = c("Objective", "n", "Mean (%)", "Median (%)",
                           "Sample SD (pp)", "Q1 (%)", "Q3 (%)"))
```

Objective	n	Mean (%)	Median (%)	Sample SD (pp)	Q1 (%)	Q3 (%)
Growth	158	10.15	10.1	4.77	7.8	12.3
Value	101	11.69	11.7	4.01	8.8	14.2

```
# Calculate Value minus Growth, rather than relying on the ordering of table rows.
mean_gap_pp <- mean(funds$return_5yr_pct[funds$objective == "Value"]) -
               mean(funds$return_5yr_pct[funds$objective == "Growth"])
mean_gap_pp
```

```
[1] 1.545325
```

- The mean five-year annualized return is about **10.15% for Growth** and **11.69% for Value**, a difference of **1.55 percentage points** in this list.
- Medians are 10.1% and 11.7%.
- The sample SDs, about 4.77 and 4.01 percentage points, describe **variation across the listed funds' five-year annualized returns**.
- They do not measure the volatility of an individual fund over time.
- The overlapping distributions matter: these summaries do not identify a universal winner.
- They also give each fund equal weight; the mean is not the return on a portfolio weighted by money invested.

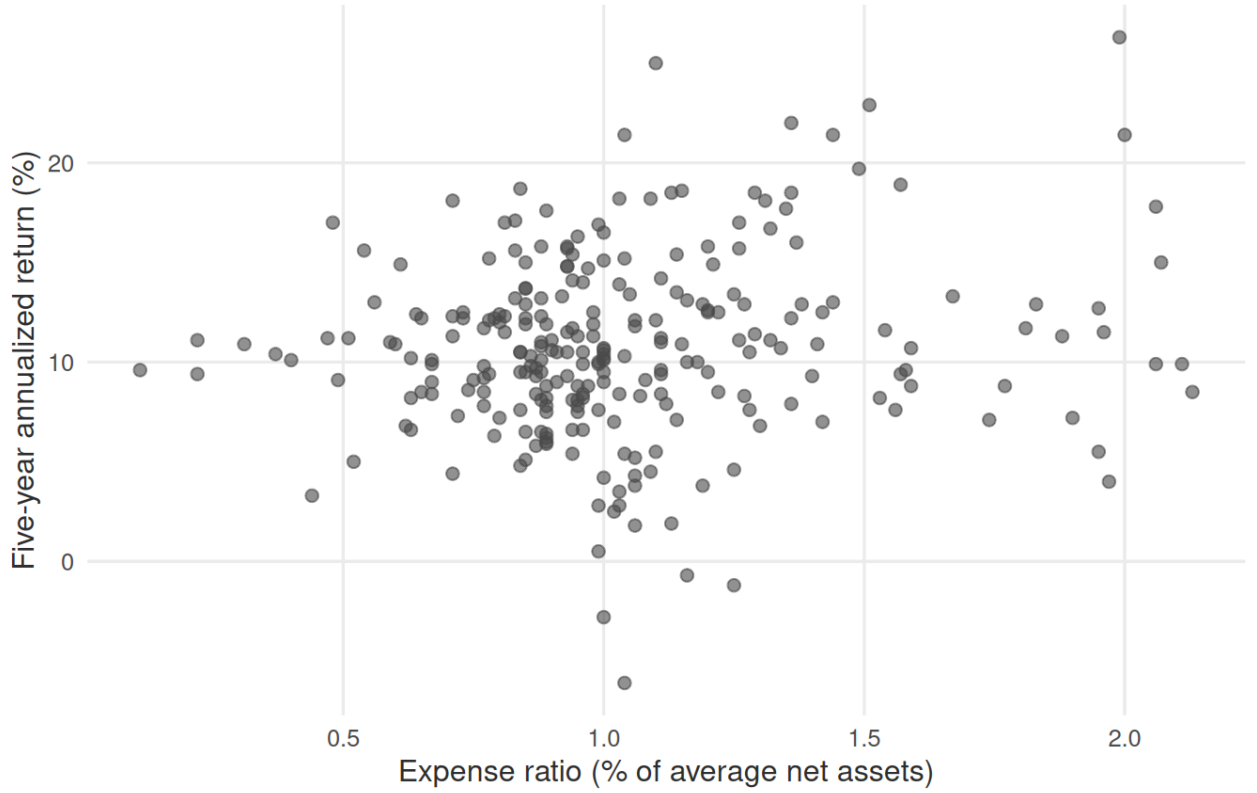
Your turn A

- Repeat the grouped summary using `return_2001_pct`.
- Compare the size and direction of the gap with the five-year comparison.
- Explain why changing the observation period changes the business question.

5. Inspect a numerical relationship

Question: Do higher expense ratios tend to occur with higher or lower five-year annualized returns in these records?

```
# A scatterplot retains individual observations; transparency shows overlapping
        points.
ggplot(funds, aes(x = expense_ratio_pct, y = return_5yr_pct)) +
  geom_point(colour = petra_ink, alpha = 0.6, size = 2) +
  labs(x = "Expense ratio (% of average net assets)",
       y = "Five-year annualized return (%)")
```



Expense ratios and historical five-year annualized returns. Each point is one fund.

- Describe direction, form, strength and unusual points before calculating a coefficient.
- The expense ratio is a single reported characteristic; the file does not give its complete five-year history.
- This plot cannot isolate the effect of changing expenses.

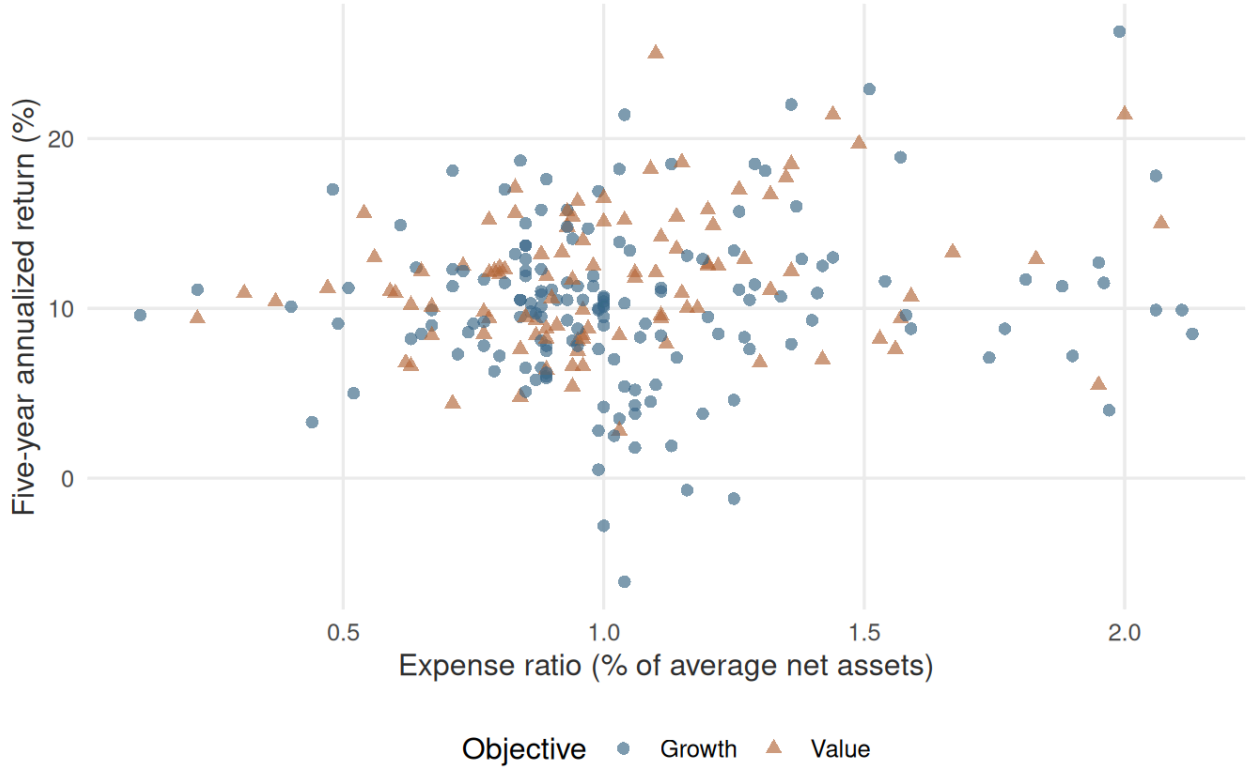
```
# cor() calculates Pearson's linear correlation using complete pairs.
complete_pairs <- complete.cases(funds[c("expense_ratio_pct", "return_5yr_pct")])
relationship <- data.frame(
  Complete_pairs = sum(complete_pairs),
  Pearson_r = cor(funds$expense_ratio_pct[complete_pairs],
                  funds$return_5yr_pct[complete_pairs])
)
knitr::kable(relationship, digits = 3)
```

Complete_pairs	Pearson_r
259	0.142

- Pearson's **r is approximately 0.142**, a weak positive linear association in this list.
- It is unitless and summarizes a linear pattern.
- It does not mean a 14.2% return, a 14.2-percentage-point increase, or that expenses explain 14.2% of return variation.
- A small correlation also does not exclude a nonlinear relationship.

6. Examine the groups within the relationship

```
ggplot(funds, aes(x = expense_ratio_pct, y = return_5yr_pct,
                  colour = objective, shape = objective)) +
  geom_point(alpha = 0.65, size = 2) +
  scale_colour_manual(values = petra_groups) +
  labs(x = "Expense ratio (% of average net assets)",
       y = "Five-year annualized return (%)", colour = "Objective", shape =
         "Objective")
```



The same relationship, identifying Growth and Value funds.

Your turn B

- Calculate the expense–return correlation separately for Growth and Value funds.
- Compare the results with the overall correlation.
- Explain why an overall relationship can hide differences between groups.
- Use the plot as evidence, rather than treating the coefficient as a complete explanation.

```
# Complete the expressions using the variables introduced above.
funds |>
  group_by(objective) |>
  summarise(n = n(), r = cor(..., ...), .groups = "drop")
```

7. Evaluate the claims

Discuss what is supported, what is uncertain and what additional evidence would be needed.

1. Value funds had a higher mean five-year annualized return in this historical list.
2. Every Value fund outperformed every Growth fund.
3. A larger cross-fund SD proves that each fund fluctuated more from year to year.
4. Raising a fund's expense ratio would improve its return because the overall correlation is positive.
5. Correlating 2001 returns with 1997–2001 returns gives independent evidence of performance persistence.

- For claim 5, note that **2001 is included in the five-year period**.
- An association between overlapping return windows is not evidence about independent periods.
- There is also no random assignment of fund expenses or objectives, and the source does not establish probability selection.
- Significance tests, confidence intervals and regression inference belong to Part 3.
- Before extending these analyses, we would need to justify the target population, selection process, independence and model assumptions.
- Fund returns can be related through common holdings and market conditions.

8. Independent analyst briefing

Produce a **one-page descriptive briefing**, supported by your code and output:

1. Compare five-year annualized returns across the ordered `risk` categories using a boxplot and a summary table with group sizes, medians and IQRs.
2. Use the expense–return plot and correlations to explain whether the numerical association is consistent across objectives.
3. State three findings, including at least one in percentage points, and two limits on interpretation.
4. Identify one additional variable or study-design detail needed before making a stronger business recommendation.

- Make your choices and explain them.
- A descriptive difference alone is not a recommendation to buy a fund or a forecast of future performance.

Source and data notes

- Berenson, Levine and Krehbiel, *Basic Business Statistics: Concepts and Applications*, ninth edition, **FUNDS2002** companion data.
- Printed p. 40 defines the 1997–2001 comparison; p. 75 and team projects pp. 130–131 define variables and related comparisons.
- The course questions adapt these data to Part 2; they do not reproduce every original exercise.
- Source order and all numerical observations are preserved.
- Column names were simplified and spaces in `Type` trimmed.
- Returns are historical; original collection and probability-selection methods remain unverified.
- Full data dictionary.